

From detector signals to $\Delta(x, Q^2)$: the reconstruction chain of the inclusive and coherent $\cos 2\varphi$ measurements

2026-08-24 · companion to the $\cos 2\varphi$ projection report (money plots 5–7) and to plans/07 WP3 · quantitative figures from the new `polligen.reco` module (15 tests) · every number below is reproducible with `evgen/scripts/reco_chain_figures.py`

The projection report presents the double-helicity-flip structure function $\Delta(x, Q^2)$ as φ' pseudo-data and as extracted $x\Delta$ points. This note asks the experimentalist's question behind those plots: what does the detector actually record, event by event, in the inclusive and in the coherent intact-⁶Li measurement; how is the azimuth that carries the observable built from the reconstructed four-vectors and the known spin direction; which kinematic reconstruction assigns (x, Q^2) to an event; which estimator turns spin-sorted φ' histograms into an amplitude; and how the amplitude becomes Δ . It then audits, step by step, which of these operations the simulation performs. The audit's conclusion is that the generator-level machinery is complete and validated but that the *measurement* layer — four-vectors, frames, kinematic reconstruction, spin-state sorting, the Roman-Pot measurement — was absent; a seed of that layer now exists, and three of its first results change the analysis design.

Three findings that change the analysis. (1) Three of the four inclusive sweet-spot bins of money plot 5 sit at $y = 0.010$ – 0.025 , where the scattered electron carries 98–99% of the beam energy and the electron method gives $\delta y/y = 50$ – 120% even with a 1.2% calorimeter: x must come from the hadronic final state (Σ or Jacquet–Blondel y), which the simulation does not generate. With a 15–20% hadronic- y resolution the super-bins keep 75–83% purity and the amplitude in the reconstructed bin is within 1–4% of the true-bin value. (2) The single-fill $\cos 2\varphi'$ fit — the estimator of money plots 5–7 — measures the detector's own $\cos 2\varphi'$ acceptance harmonic divided by P_{zz} : a 3% harmonic aligned with the vertical spin axis (the natural symmetry axis of both the crossing angle and the Roman-Pot cutout) fakes an amplitude four times the signal. The spin-state-sorted ratio of $m = \pm 1$ -rich and $m = 0$ -rich fills cancels any acceptance bin by bin, turns relative-luminosity errors into a constant, and is $1.5\times$ *more* precise than the single fill. (3) In the coherent channel the anchored deformation term modulates the *recoil* azimuth $\varphi_t - \varphi_s$, not the electron azimuth used in the projection; the Roman-Pot cutout fakes $\langle \cos 2\varphi_t \rangle \approx 0.5$ for a 25% aspect ratio (against a physics $a_2 \approx 0.036$), and the near-beam cut is angular, so the tag acceptance is 67% / 9% / 10^{-8} at 20.5 / 50 / 137.5 GeV/u for the same optics.

1 • What is measured

Both measurements share the trigger object — a scattered electron in the central detector — and the spin bookkeeping of the ion beam. The coherent measurement adds the intact ⁶Li in the far-forward spectrometers and an exclusivity selection. The observable in both cases is not a cross section but a *Fourier coefficient of a ratio* of spin-sorted azimuthal distributions; Δ follows from that coefficient through kinematic factors and the unpolarized structure functions of the same data set.

RECORDED PER EVENT	INCLUSIVE E ⁶ LI → E' X	COHERENT E ⁶ LI → E' X ⁶ LI(G.S.)
Beam and spin	Bunch-crossing ID → spin state of the ion bunch (source pattern: $m = \pm 1$ -rich, $m = 0$ -rich); tensor polarization P_{zz} per fill from the polarimeter (scale $\approx 3\%$); alignment axis $\hat{n}$ = stable spin direction (vertical at IP6) with its measured tilt; luminosity per spin state.	
Central detector	e' candidate: track (p, θ, φ at the vertex, charge) matched to an EMCal cluster (E , position); E/p and shower shape for electron ID; the hadronic final state (all other tracks and clusters) for $\Sigma = \Sigma_h(E_h - p_{z,h})$; vertex.	The same, plus the diffractive system X (its mass M_X and the rapidity gap) — the only handle on x_p , since the recoil momentum is unresolved.

Far forward	Not required (breakup fragments of ${}^6\text{Li}$ are mostly beam-blind at IP6; plans/00).	Roman-Pot track at beam rigidity ($ R - 1 < 0.05$) with transverse displacement above the 10σ envelope $\rightarrow (\theta_x, \theta_y)$ at the IP via the optics transport $\rightarrow p_T, \varphi_T, t$; vetoes in OMD ($p, {}^3\text{He}$), ZDC (n), B0 EMCal (de-excitation γ); Z-identification if available (open question #19).
Observable	$\cos 2\varphi'$ coefficient of the ratio of spin-sorted $N_f(\varphi')$ in (x, Q^2) bins, $\varphi' = \varphi_e - \varphi_S$	$\cos 2\alpha, \cos 2\beta$ and $\cos(\alpha+\beta)$ coefficients of the same ratio in (x_p, t, Q^2) bins, $\alpha = \varphi_e - \varphi_S, \beta = \varphi_t - \varphi_S$

2 • Angles and frames

For an unpolarized electron on a spin-1 target whose alignment axis has polar angle θ_S and azimuth φ relative to the virtual photon and the lepton plane, the Hoodbhoy–Jaffe–Manohar cross section for the projection state m reads [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\varphi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3}a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3}a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_S \Delta(x, Q^2) \cos 2\varphi \right], \quad c_m = 3m^2 - 2$$

and a fill with populations p_m along a transverse axis ($\theta_S = 90^\circ$) has the φ' distribution $1 + a_2 \cos 2\varphi'$ with

$$a_2 = -P_{zz} \frac{1-y}{y^2} \frac{\Delta}{D_\varphi}, \quad D_\varphi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{A} \equiv \frac{a_2}{P_{zz}}, \quad P_{zz} = \sum_m p_m c_m.$$

Which angle. The φ of the master formula is defined in a frame in which the target momentum P and the virtual photon q are collinear; it is the azimuth of the transverse spin about q measured from the lepton plane. The covariant definition used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4] applies verbatim to the spin four-vector S (and, with P' in place of S , to the recoil azimuth of §4):

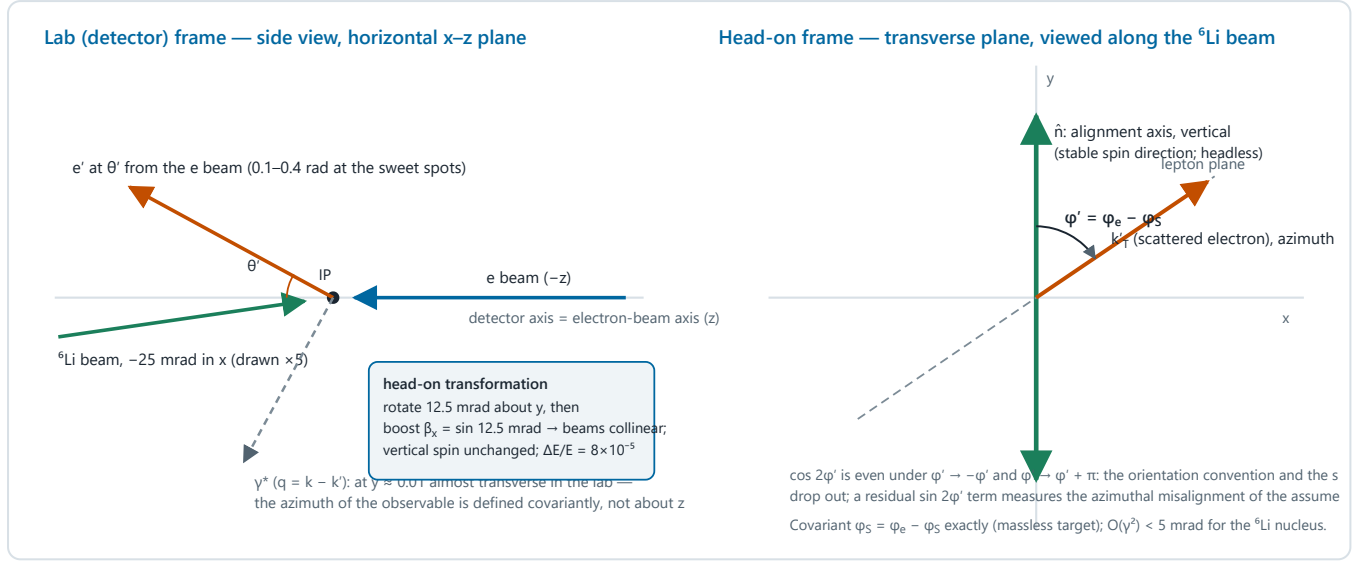
$$\cos \varphi_S = -\frac{l_\perp \cdot S_\perp}{|l_\perp| |S_\perp|}, \quad \sin \varphi_S = -\frac{\epsilon_{\mu\nu\rho\sigma} l^\mu S^\nu P^\rho q^\sigma}{(P \cdot q) \sqrt{1+\gamma^2} |l_\perp| |S_\perp|}, \quad v_\perp^\mu = g_\perp^{\mu\nu} v_\nu, \quad \gamma^2 = \frac{M^2 Q^2}{(P \cdot q)^2}$$

with $g_\perp$ the projector orthogonal to P and q [3]. Evaluated in the head-on frame — ion along $+z$, electron along $-z$, the frame of the whole fast simulation — with a spin axis transverse to the beam at lab azimuth φ_S and the scattered electron at lab azimuth φ_e , this reduces *exactly* to $\varphi' = \varphi_e - \varphi_S$ for a massless target, and to the same with corrections of order γ^2 (below 5 mrad for $\gamma^2 \leq 0.02$, i.e. everywhere in the sweet spots) for the massive nucleus (`reco.azimuth_wrt_lepton_plane`, verified against an explicit boost-and-rotate construction of the collinear frame to 10^{-13}). The lab-angle shortcut is therefore legitimate for the inclusive measurement; the analysis should nevertheless evaluate the covariant formula from the reconstructed four-vectors, because it is exact, costs nothing, and is the same code that the coherent channel needs.

Which frame. The beams cross at 25 mrad in the horizontal plane at IP6. The ePIC axis coincides with the electron beam, and the ion beam enters at -25 mrad in x (the convention of the far-forward gun scan in `tools/fullsim`). The head-on frame is reached by a rotation of 12.5 mrad about y followed by a boost $\beta_x = \sin 12.5$ mrad; energies change by 8×10^{-5} and the vertical spin direction is untouched. For the scattered electron the boost and the rotation cancel at first order because the detector axis is the electron axis: the residual distortion of its azimuth about the detector axis is an odd-harmonic effect of 0.7×10^{-3} ($\theta' = 107$ mrad) to 2.4×10^{-3} (383 mrad) in $\cos \varphi$, with the $\cos 2\varphi$ harmonic untouched to 10^{-17} and the response of a true $\cos 2\varphi$ term equal to one within 10^{-5} . The transformation is applied anyway (it is the standard first step of every ePIC inclusive analysis and it is exact), but the crossing angle is not a source of a fake $\cos 2\varphi'$ for the electron. It is one for the recoil (§4).

Orientation and the axis. The tensor alignment axis is headless — the $m = +1$ and $m = -1$ populations enter through $c_m = 3m^2 - 2$ symmetrically — and $\cos 2\varphi'$ is even under $\varphi' \rightarrow -\varphi'$ and $\varphi' \rightarrow \varphi' + \pi$, so neither the orientation convention of φ nor the sign of $\hat{n}$ can change the result. What matters is the axis direction modulo π : a misalignment δ of the assumed azimuth rotates the modulation, $\cos 2(\varphi' - \delta) = \cos 2\delta \cos 2\varphi' + \sin 2\delta \sin 2\varphi'$, so the fit should carry a $\sin 2\varphi'$ term; since the only physics harmonic in the tensor ratio is $\cos 2\varphi'$ (the m -

symmetric fills set $a_1 = 0$, and the b_1 -type constant is φ' -independent), the $\sin 2\varphi'$ coefficient measures δ from the data themselves — a self-calibration of the spin axis at the precision of the modulation itself. A polar tilt ε of the axis out of the transverse plane dilutes by $\sin^2\theta_S = 1 - \varepsilon^2$ (0.8% at 5°).



Schematic 1 — frames and the azimuth of the observable. Left: the detector frame with the 25 mrad crossing angle; the scattered electron is measured at (E', θ', φ_e) about the electron-beam axis, and the virtual photon is far from the beam axis at low y . Right: in the head-on frame the alignment axis is vertical and the physics azimuth is the angle between the lepton plane and the axis, which the covariant formula reproduces as $\varphi_e - \varphi_S$.

3 • The inclusive chain

- 1. Event selection and electron identification.** One e' candidate with a track matched to an EMC cluster; E/p and shower-shape cuts against π^- (photoproduction) and a charge-symmetric subtraction (wrong-sign tracks, positron running) for pair-symmetric background at high y . Efficiency $\varepsilon_{\text{elD}}(\eta) = 0.70 - 0.95$ in the ATHENA/ECCE anchors used by the money_delta study; no official ePIC curve exists.
- 2. Scattered-electron kinematics.** E' from the calorimeter ($2\%/\sqrt{E} \oplus 1\%$, i.e. 1.2% at $E' \approx E_e$) or the tracker (3% at $\eta \approx -3$ in the tracking-only table), θ' and φ_e from the track at the vertex (1–3 mrad), transformed to the head-on frame. The four-vectors k, k', P and the spin four-vector $S = (0, \hat{n})$ enter the covariant azimuth.
- 3. Kinematic reconstruction.** The electron method,

$$Q_e^2 = 4E_e E' \sin^2 \frac{\theta'}{2}, \quad y_e = 1 - \frac{E'}{E_e} \cos^2 \frac{\theta'}{2}, \quad x_e = \frac{Q_e^2}{s y_e}, \quad \frac{\delta y_e}{y_e} = \frac{1-y}{y} \left[\frac{\delta E'}{E'} \oplus \tan \frac{\theta'}{2} \delta \theta' \oplus \frac{\delta E_e}{E_e} \right]$$

has an excellent Q^2 ($\delta Q^2/Q^2 \approx \delta E'/E' \oplus \cot(\theta'/2) \delta \theta' \approx 1-2\%$) but loses y as $(1-y)/y$: at $y = 0.01$ a 1.2% energy resolution gives $\delta y/y = 120\%$, and the electron-ring energy spread alone ($\approx 10^{-3}$) gives 10%. Table 1 shows where money plot 5 sits. At the mid-energy configuration three of the four sweet spots are at $y = 0.010-0.025$, the scattered electron carries 98–99% of the beam energy at $\eta \approx -2.9$ to -2.4 , and x is not measurable from e' alone; at the top energy all four spots sit at $y \approx 0.01$. The remedy is the one HERA used at low y [5,6]: y from the hadronic final state,

$$y_{JB} = \frac{\Sigma}{2E_e}, \quad y_{\Sigma} = \frac{\Sigma}{\Sigma + E' (1 - \cos \theta_e)}, \quad \Sigma = \sum_{h \neq e'} (E_h - p_{z,h}), \quad x = \frac{Q_e^2}{s y_{\Sigma}} \text{ (mixed method)}$$

with θ_e measured from the ion direction. The Σ method is insensitive to collinear initial-state radiation, and both are insensitive to particles lost along the ion beam ($E - p_z \approx 0$ for forward remnants). Their resolution is set by hadronic calorimetry and by losses in the electron direction; the Yellow-Report-era studies reach $y \approx 0.01$

with $\delta y/y$ of order 15–30%, which is the band assumed here. The low-energy configuration moves the same (x , Q^2) to $y = 0.03$ – 0.2 (Table 1), so the energy scan is itself part of the reconstruction strategy for the $x \approx 0.1$ bins.

SWEET SPOT	X, Q^2 [GeV ²]	$E(5) \times {}^6\text{Li}(20.5): Y$	$E(10) \times {}^6\text{Li}(50): Y, E' \text{ [GeV], } \Theta' \text{ [mrad], } H$	$E(18) \times {}^6\text{Li}(137.5): Y$	$\Delta Y/Y, E' \text{ ALONE (1.2\%)}$
1	0.056, 1.14	0.031	0.010, 9.93, 107, -2.93	0.011 ($x = 0.014$)	1.2
2	0.022, 1.14	0.078	0.025, 9.78, 108, -2.92	0.026 ($x = 0.0056$)	0.47
3	0.141, 3.13	0.054	0.011, 9.97, 177, -2.42	0.011 ($x = 0.028$)	1.1
4	0.141, 14.3	0.196	0.051, 9.85, 383, -1.64	0.010	0.23

Table 1 — the four sweet-spot super-bins of money plot 5 (selected per Q^2 band from the significance map at $e(10) \times {}^6\text{Li}(50)$) and where the same bins sit at the other two energies; the top-energy spots are chosen by the same rule and land at different x .

4. **The azimuth.** $\phi' = \phi_e - \phi_s$ from the covariant formula (§2), with ϕ_s the alignment-axis azimuth of that bunch from the spin bookkeeping. The angular resolution of the track (1–3 mrad) dilutes $\cos 2\phi'$ by $\exp(-2\sigma_{\phi'}^2) \approx 1 - 2 \times 10^{-5}$: negligible.
5. **Binning in reconstructed (x, Q^2).** Figure 1(c) quantifies the migration for the mixed method with the sampler's own events: with $\delta y_{\text{had}}/y = 15\%$ (20%, 30%) the 3×3 super-bins keep a purity of 0.81–0.83 (0.74–0.77, 0.61–0.65) and the amplitude seen in the reconstructed bin is 0.98–0.99 (0.96–0.99, 0.92–0.98) of the true-bin value, because $A(x)$ varies slowly across the bin. With the electron method alone the low- y spots collapse to purities of 0.41–0.56 with 26–53% of the events surviving in the bin. The standard treatment — a migration matrix from the same simulation and either an unfolding or model bin-centering — is the same as for any structure-function measurement; what is new is only that the correction must be built on the hadronic-method resolution.
6. **Spin-state sorting and the estimator.** The events of each (x, Q^2) bin are histogrammed in ϕ' separately for each spin state f of the bunch pattern, $N_f(\phi')$. Any smooth ϕ' -dependent efficiency $\varepsilon(\phi')$ multiplies every N_f identically, so the bin-wise ratio

$$R(\phi') = \frac{\sum_f w_f Y_f(\phi')}{\sum_f Y_f(\phi')}, \quad Y_f = \frac{N_f}{L_f}, \quad w_f = P_f - \bar{P}, \quad R(\phi') \simeq \sigma_P^2 [\kappa + \hat{A} \cos 2\phi'], \quad \sigma_P^2 = \frac{\sum_f L_f (P_f - \bar{P})^2}{\sum_f L_f}$$

is free of $\varepsilon(\phi')$ exactly, and a relative-luminosity error δ_f enters it only as the ϕ' -independent constant $\sum w_f \delta_f / F$, orthogonal to the harmonic (second order: $\delta \times \hat{A} \approx 10^{-6}$). A weighted least-squares fit of $R(\phi') = c + d \cos 2\phi' (+ e \sin 2\phi')$ gives $\hat{A} = d/\sigma_P^2$ after the finite-bin dilution $\sin w/w$ is divided out, and $\kappa = c/\sigma_P^2$ is the transverse tensor asymmetry of the b_1 sector, a by-product. For two fill types with equal luminosity the statistical error is

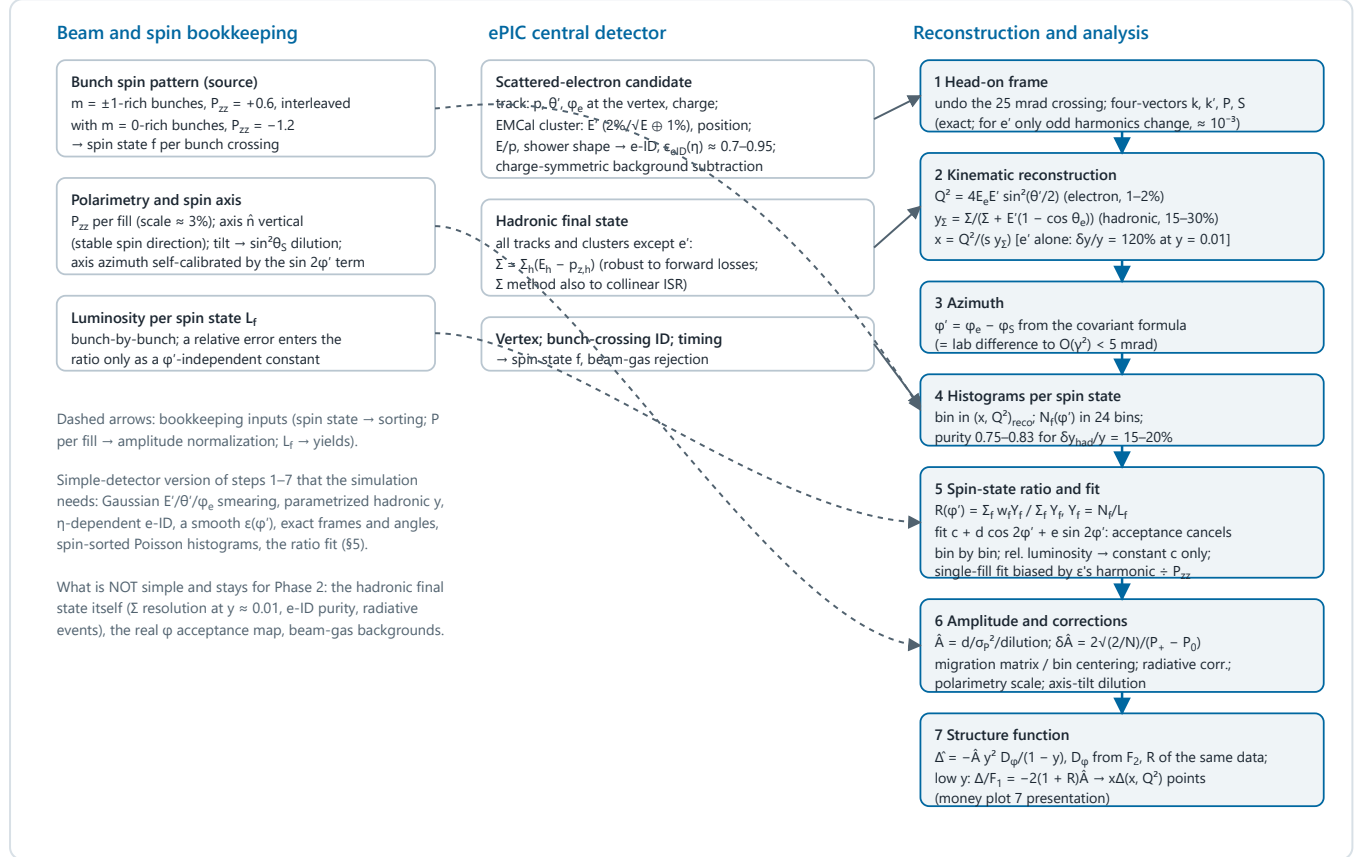
$$\delta \hat{A} = \frac{2}{P_+ - P_0} \sqrt{\frac{2}{N}} \quad (\text{two fills}) \quad \text{vs} \quad \delta \hat{A} = \frac{1}{P_{zz}} \sqrt{\frac{2}{N}} \quad (\text{single fill})$$

so that $m = \pm 1$ -rich bunches at $P_+ = +0.6$ alternated with $m = 0$ -rich bunches at $P_0 = -1.2$ (the same source purity gives $-2P_+$, the pattern of the A_{zz} run plan) are $1.5 \times$ more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity; an unpolarized reference fill ($P_0 = 0$) would instead cost a factor two. Figure 1(d) shows the pseudo-experiments: under an efficiency $\varepsilon = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ the single-fill fit returns $\hat{A} = 0.062$ for a true 0.0118 — the acceptance harmonic divided by P_{zz} — while the ratio is unbiased with the analytic error. The projection report's statement that the modulation is "self-normalized within a single fill" holds for relative luminosity but not for acceptance; since a percent-level $\cos 2\phi$ harmonic aligned with the vertical axis is expected (the crossing-angle plane is horizontal, module boundaries and dead areas are not azimuthally random, and the Roman-Pot cutout is a rectangle), the spin-state-sorted ratio is the estimator of the measurement.

7. **Amplitude to Δ .** With F_2 and $R = \sigma_L/\sigma_T$ of ${}^6\text{Li}$ per nucleon measured in the same data (the unpolarized cross section of the same sample, with its own radiative corrections) or taken from nuclear-PDF fits,

$$\hat{\Delta}(x, Q^2) = -\hat{A} \frac{y^2 D_\phi}{1-y} \rightarrow -\hat{A} \frac{F_2}{x} = -2(1+R) F_1 \hat{A} \quad (y \ll 1), \quad \text{i.e.} \quad \frac{\Delta}{F_1} = -2(1+R) \hat{A}$$

At the sweet spots the conversion is therefore insensitive to y — the $(1-y)/y^2$ factor that makes low y attractive for the amplitude cancels against D_ϕ — and Δ/F_1 is essentially a direct ratio measurement; the y mis-measurement of the electron method affects the x label of the point, not the amplitude. Remaining corrections: the tensor rate shift $(2/3)a_m(b_1 + \dots)$ in D_ϕ , at the 10^{-3} level and measured by κ ; target-mass terms of order y^2 ; radiative corrections, uncalculated for tensor observables (plans/o7 WP4); the per-nucleon and dilution conventions, which are interpretation rather than reconstruction; and the polarimetry scale $\delta P_{ZZ}/P_{ZZ} \approx 3\%$, common with A_{ZZ} .



Schematic 2 — the inclusive reconstruction chain. Left and middle: what is recorded; right: the seven reconstruction and analysis steps of §3 with the quantity each produces and its dominant limitation. The measurement's observable is the $\cos 2\phi'$ coefficient of the spin-state ratio (step 5), not the modulation of a single histogram.

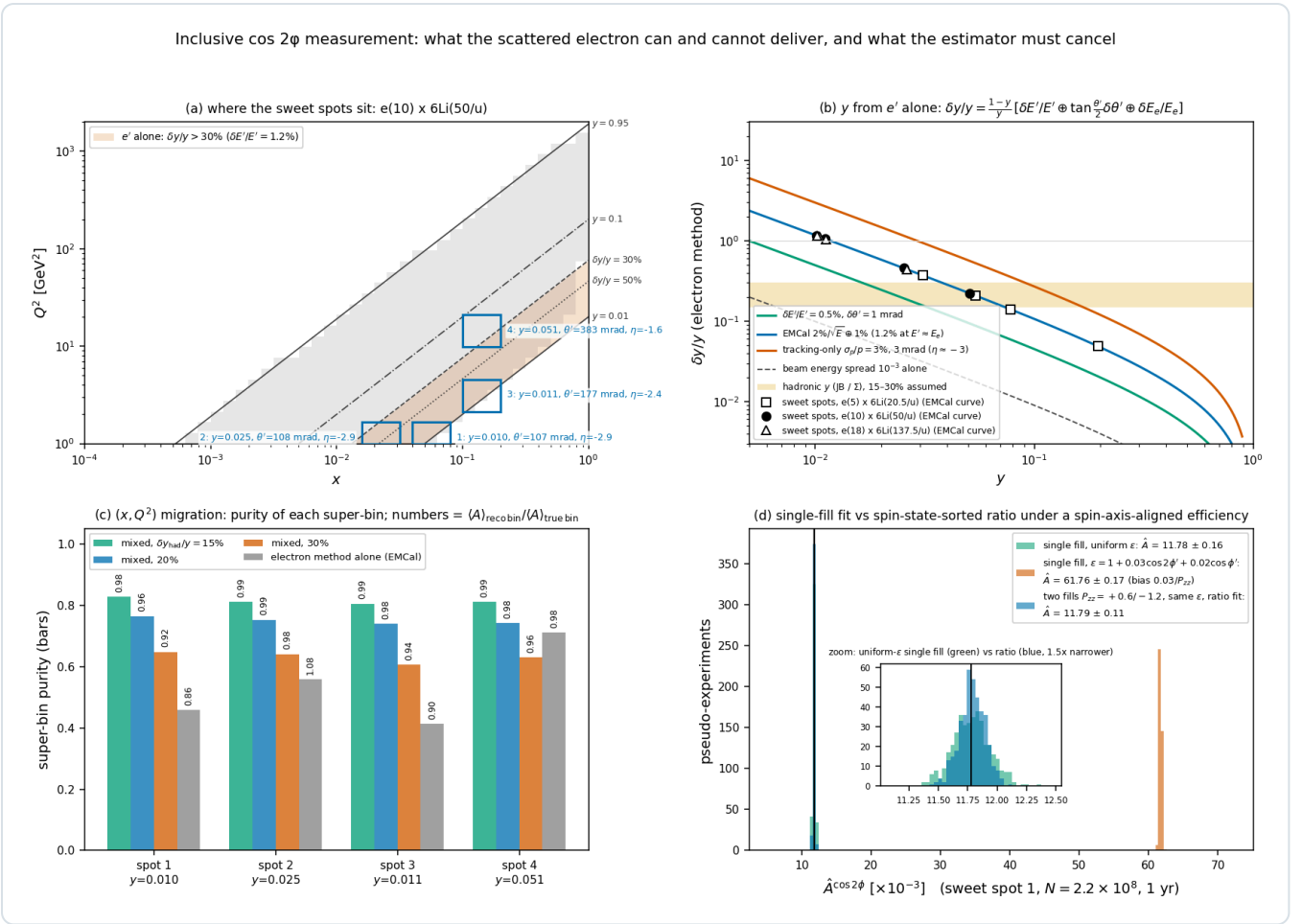


Figure 1 — the inclusive measurement quantified ($e(10) \times 6\text{Li}50$, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso- y lines and the four sweet-spot super-bins of money plot 5; the shaded band is where the electron alone gives $\delta y/y > 30\%$ for a 1.2% energy resolution: spots 1–3 lie inside it, spot 4 just above. **(b)** The electron-method $\delta y/y$ versus y for calorimeter, tracking, and beam-energy-spread inputs, with the sweet spots of the three energies placed on the calorimeter curve and the assumed 15–30% hadronic-method band. **(c)** Purity of each super-bin after smearing the sampler's events with the mixed method for three hadronic- y resolutions and with the electron method alone; the numbers give the ratio of the amplitude averaged over the events landing in the reconstructed bin to that of the true bin. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at the 1-year statistics: the single-fill fit with a uniform efficiency (green) recovers the truth, the same fit under $\epsilon = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ (vermillion) is biased by $0.03/P_{zz}$, and the two-fill ratio fit under the same efficiency (blue) is unbiased and $1.5\times$ more precise (inset).

4 • The coherent chain

The coherent measurement keeps steps 1–7 for the electron and adds the recoil. Its kinematics is fixed by two facts of the projection report: the recoil keeps the beam rigidity ($R = 1.000$ for $A/Z = 2$, and the momentum loss $x_p < 0.01$ is below the Roman-Pot momentum resolution), so the only measured recoil quantity is its transverse displacement from the beam orbit; and coherence confines the sample to $x \lesssim 0.01$, i.e. to $y \gtrsim 0.05$ at the mid energy (median $x = 3.5 \times 10^{-3}$, $y \approx 0.2$), where the electron method is adequate and the low- y problem of §3 does not arise.

- Roman-Pot track.** Hits in the two stations (AC-LGAD, ≈ 30 ps timing for bunch association) form a local track; the optics transfer matrices referenced to the beam orbit convert it to the angle pair (θ_x, θ_y) at the IP relative to the ion axis, so the crossing angle and the reference orbit are removed by construction, and the vertex from the central detector supplies the position term. The momentum fraction x_L is set to 1 (unresolved), hence $p_T = A p_u \sqrt{(\theta_x^2 + \theta_y^2)}$ and $\phi_t = \text{atan2}(\theta_y, \theta_x)$. The resolution is dominated by the angular divergence of the bunch itself, $\delta p_T = A p_u \sigma_\theta = 22$ MeV at 50 GeV/u for $\sigma_\theta = 73$ μrad , giving $\sigma(\phi_t) = 0.09$ rad at $p_T = 0.25$ GeV and a $\cos 2\phi_t$ dilution of 1.5% (Figure 2d).
- Momentum transfer.** Exactly, with $x_L = 1 - x_p$ the plus-momentum fraction,

$$t = -\frac{p_T^2 + M^2 x_P^2}{1 - x_P} \simeq -p_T^2, \quad t_{\min} = \frac{M^2 x_P^2}{1 - x_P} = 3.2 \times 10^{-3} \text{ GeV}^2 \quad (x_P = 0.01), \quad x_P = \frac{M_X^2 + Q^2 - t}{W^2 + Q^2 - M^2}$$

so the Roman Pots give t , and x_P (or $\beta = x_A/x_P$) must come from the diffractive mass M_X of the central detector — a coarse variable (a few x_P bins), sufficient for the x_P -dependence test of the two-component fit. The covariant recoil azimuth differs from the lab difference $\varphi_e - \varphi_t$ by less than 0.2 mrad at these kinematics, so here too the lab shortcut is safe.

3. Exclusivity selection. A beam-rigidity Roman-Pot track above the envelope; no OMD track (p at $R = 0.5$, ^3He at 0.75 in the main window), no ZDC neutron, no Bo EMCal photon (the 3.56 MeV state's de-excitation γ boosted to 0.1–0.4 GeV); a rapidity gap between X and the beam; Z -identification of the track if the pots resolve $Z = 3$ from 2 and 1 (open question #19), otherwise a two-component fit of the $|t|$ spectrum against the ten-times-flatter $\alpha + d$ breakup slope; timing and vertex association against beam-gas coincidences. Because the incoherent cross section is m -state-blind, residual breakup dilutes the tensor ratio without faking it; its fraction is measured in the same data from the $|t|$ fit.

4. Three azimuths, two harmonics. With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ (lepton plane to axis) and $\beta = \varphi_t - \varphi_S$ (recoil to axis), and $\alpha - \beta = \varphi_e - \varphi_t$ is the lepton-plane–recoil angle of unpolarized diffractive DIS. The rank-2 tensor polarization enters only through $e^{\pm 2i\varphi_S}$, so at leading power in the momentum transfer the spin-sorted distribution has the structure

$$N_f(\alpha, \beta) \propto \varepsilon(\alpha, \beta) [1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_f(\kappa + a_e \cos 2\alpha + a_t \cos 2\beta + a_m \cos(\alpha + \beta))]$$

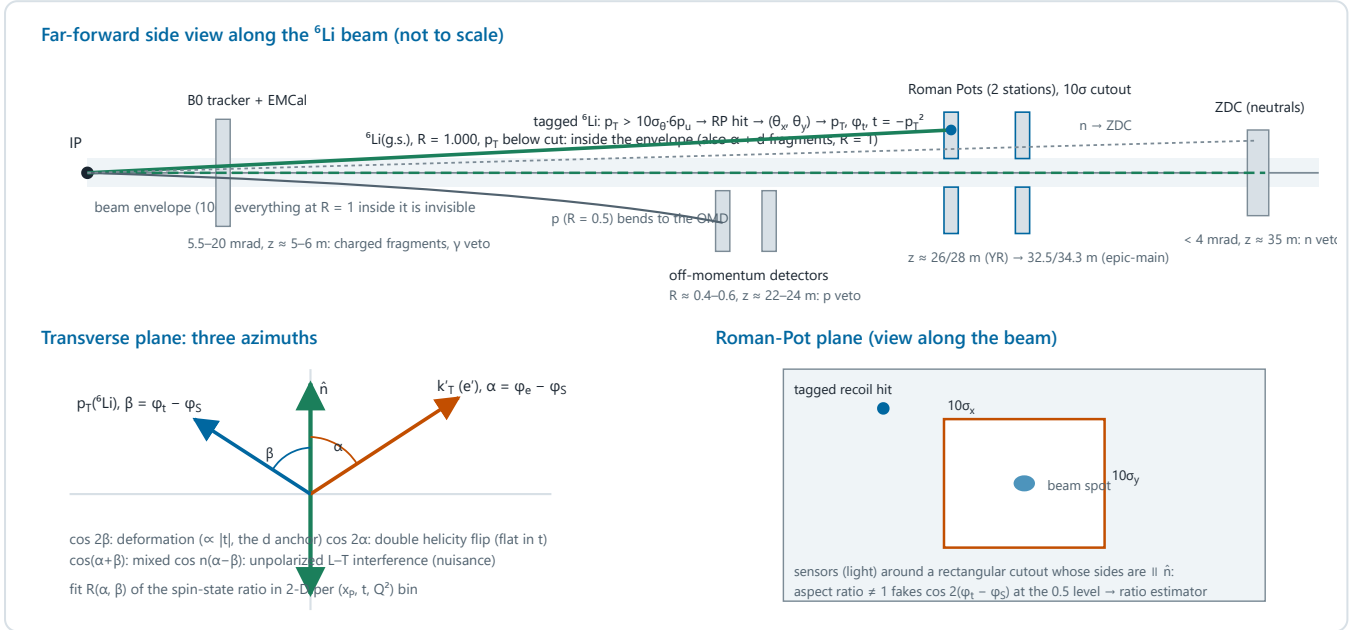
The deformation term of the polarized-deuteron anchor — the modulation of the diffractive slope by the aligned gluon density, $a_2(t) \propto |t|$, digitized in `coherent.MANTYSAARI_A2_DEUTERON` — is a_t : it lives in the *recoil* azimuth β , requires no photon helicity flip, and vanishes as $t \rightarrow 0$. The double-helicity-flip term — the coherent counterpart of Δ , an interference of the two transverse photon helicities that needs the linear polarization of the photon and hence the lepton plane — is a_e : it lives in α and is flat in t at leading power. The two mechanisms of the projection report are thus separated not only by their t -dependence but by the azimuth they modulate, which is a stronger discriminant; the mixed term a_m and the unpolarized $u_{1,2}$ are nuisance harmonics. The money-plot-6 pseudo-data place the anchored $\langle a_2 \rangle$ on an axis labelled $\varphi' = \varphi - \varphi_S$ of the electron, which is the α axis: numerically harmless for a statistical projection, but the measurement must histogram (α, β) in two dimensions per (x_P, t, Q^2) bin and extract a_t and a_e from the spin-state ratio $R(\alpha, \beta)$; in that ratio $u(\alpha - \beta)$ survives in the denominator and feeds the pure harmonics only at second order ($u_1^2 \approx 1\%$) but the mixed term at first order ($a_t u_1/2$), which fixes the fit model. (One normalization caveat to settle at the source: whether the anchor's a_n are defined with $1 + 2\Sigma a_n \cos n\Phi$ or with $1 + \Sigma a_n \cos n\Phi$ — a factor two on the digitized amplitudes that the projection code treats as the latter.)

5. Acceptance geometry and the estimator. The Roman-Pot acceptance is not a circular p_T cut. The pots sit outside the 10σ beam envelope, whose section at the pots is set by β_x, β_y and the emittances, and the sensors leave a rectangular cutout whose sides are horizontal and vertical — parallel to the spin axis. For the exponential recoil spectrum the tagged sample's azimuthal distribution is $\varepsilon(\varphi_t) = \exp[-B \rho(\varphi_t)^2]$ with ρ the cutout boundary, and its $\langle \cos 2\varphi_t \rangle$ is 0 for a square, 0.49 for an aspect ratio of 1.25 and 0.71 for 1.5 (Figure 2b) — ten to twenty times the anchored $a_t \approx 0.036$, with the same symmetry axis. A square still carries $\langle \cos 4\varphi_t \rangle = 0.34$, which mixes into $\cos 2\beta$ at second order through the physics term. No single-fill fit can survive this; the spin-state ratio of §3 cancels it exactly, bin by bin in (α, β) , and the same $m = \pm 1$ -rich / $m = 0$ -rich pattern serves both measurements.

6. The near-beam cut is angular. The documented 0.20 GeV (high-acceptance) and ≈ 0.41 –0.45 GeV (high-divergence) cuts are ten beam divergences times the 275 GeV proton momentum, $\sigma_\theta \approx 73$ and 149 μrad . The same envelope applied to a ^6Li nucleus of momentum $6p_u$ is $p_T^{\text{cut}} = 10\sigma_\theta \cdot 6p_u$: 0.09, 0.22 and

0.60 GeV at 20.5, 50 and 137.5 GeV/u, giving tag acceptances of 67%, 9% and 1.5×10^{-8} for $B = 50 \text{ GeV}^{-2}$ (Figure 2c). The projection's constant 0.20 GeV cut (13.5%) is close to the mid-energy value and far from the others; whether the lithium divergence at a given energy equals the proton's is precisely the undocumented optics question of plans/04 #11, but the scaling with $A p_u$ is kinematics, not optics, and it makes the low-energy configuration the natural home of the coherent program.

7. **Resolution and unfolding.** The divergence smearing moves the effective slope from B to $1/(1/B + 2\delta p_T^2)$ (47.7 for the numbers above) and smears $|t|$ by $2p_T\delta p_T \approx 0.01 \text{ GeV}^2$ — small against the 0.04–0.2 GeV^2 window (Figure 2d); the t -binning migration and the acceptance edge are unfolded with the same emulation, and $t_{\min}$ is a known bias of $3 \times 10^{-3} \text{ GeV}^2$ at $x_p = 0.01$ once x_p is known from M_X .



Schematic 3 — the coherent channel. Top: the far-forward measurement of the intact recoil, visible only when its transverse momentum carries it outside the beam envelope at the Roman Pots, and the vetoes that define exclusivity. Bottom left: the two physical azimuths relative to the alignment axis and the harmonic each mechanism modulates. Bottom right: the rectangular cutout of the pots, whose symmetry axes coincide with the spin axis.

Coherent intact-⁶Li channel: the recoil measurement, its acceptance geometry, and the angular near-beam cut

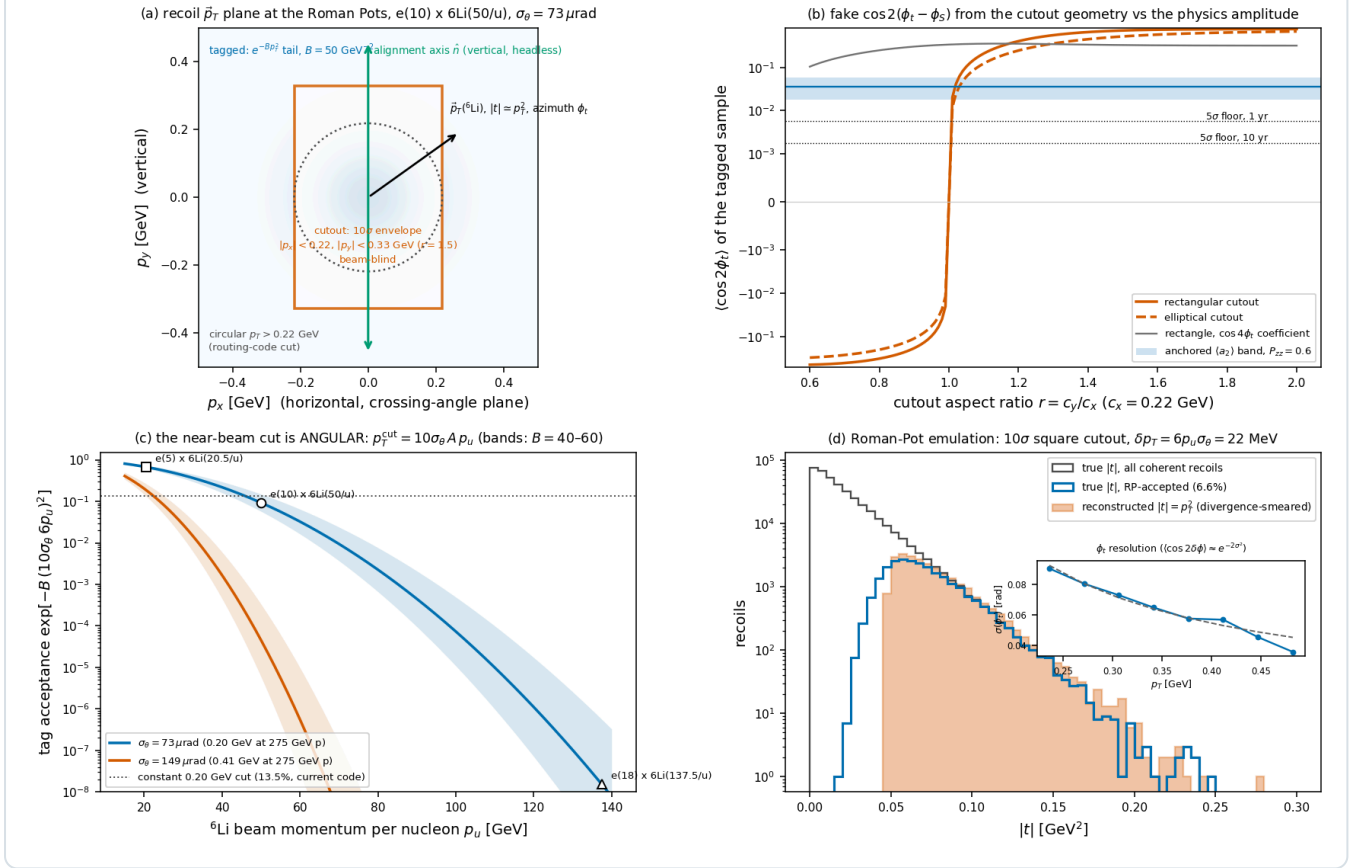


Figure 2 — the coherent measurement quantified. (a) The recoil transverse-momentum plane: the exponential coherent spectrum, the rectangular 10σ cutout (aspect ratio 1.5 drawn) against the circular cut of the routing code, and the vertical alignment axis. (b) $\langle \cos 2\phi_t \rangle$ of the tagged sample versus the cutout aspect ratio for rectangular and elliptical cutouts, with the $\cos 4\phi_t$ coefficient of the square, the anchored deformation band and the 5σ statistical floors of the 1- and 10-year programs. (c) Tag acceptance versus beam momentum per nucleon for the two proton-derived divergences, with the $B = 40\text{--}60 \text{ GeV}^{-2}$ bands, the three beam configurations, and the constant-cut value of the projection. (d) The Roman-Pot emulation (`reco.rp_measure`): true and accepted $|t|$ spectra, the reconstructed $|t| = p_T^2$ after divergence smearing, and (inset) the ϕ_t resolution against $\delta p_T/p_T$.

5 · Does the simulation do this? An audit

The table walks the chain of §§3–4 and states, for each step, what an experiment does, what the repository does, and where. "Seeded" marks the operations added in this run in `evgen/polligen/reco.py` (15 tests in `tests/test_reco.py`); they are building blocks and quantified checks, not yet a reconstructed-level pseudo-experiment.

STEP	EXPERIMENT	REPOSITORY	STATUS
Spin bookkeeping, luminosity per state, polarimetry	Bunch pattern, L_f , P_{zz} per fill with a 3% scale	<code>bookkeeping.SpinCategory / RunPlan</code> : categories with lumi shares, relative-luminosity offsets, polarimetry smearing (not exercised in the money plots)	exists
e' four-vector, lab frame, crossing angle, head-on frame	Track + cluster $\rightarrow k'$ in the lab; head-on transformation	Everything is computed in the head-on frame; <code>kinematics.scattered_electron</code> gives (E', θ, η) for acceptance only; φ_e was never constructed. Seeded: <code>reco.electron_fourvector</code> , <code>head_on_to_lab</code> , <code>lab_to_head_on</code>	partial → seeded
Electron ID and efficiency	E/p , shower shape, charge-symmetric subtraction	<code>money_delta_20260729.eps_eid(η)</code> , ATHENA/ECCE anchors (fastsim script-local); no purity model	partial
e' resolution	Calorimeter energy, track angles	<code>money_delta_20260729.smear_config</code> : Gaussian tracking-only σ_p/p , $\sigma_\theta = \sigma_\varphi$; φ smeared but unused. Seeded: <code>reco.smear_electron</code> , <code>emcal_resolution</code> , <code>tracking_resolution</code>	partial → seeded

Kinematic reconstruction (electron / JB / Σ / mixed)	Mixed method at low y from the hadronic final state	Electron method only (20260729); no hadronic final state anywhere in polligen. Seeded: <code>reco.electron_method</code> , <code>electron_method_resolution</code> , parametrized <code>hadronic_y</code> , <code>mixed_method</code>	missing → stand-in
The physics azimuth	Covariant ϕ_S from k, k', p, S	<code>sample.sample_state</code> draws ϕ' directly from the modulation and adds a known ϕ_S ; no reconstruction. Seeded: <code>reco.azimuth_wrt_lepton_plane</code> (verified against an explicit collinear-frame construction)	missing → seeded
Reconstructed-level (x, Q ²) binning and migration	Reco bins, migration matrix, unfolding	(x, Q ²) migration in 20260729 with the electron method, amplitude carried as a per-bin truth label; money plots 5–7 bin in true (x, Q ²). Seeded: the migration study of Figure 1(c)	partial
Spin-state-sorted ratio estimator for $\cos 2\phi'$	R(ϕ') of m = ± 1 -rich vs m = 0-rich fills; fit with $\sin 2\phi'$	<code>transverse_tensor_plan</code> is a single category; <code>estimators.cos2phi_fit_binned</code> fits one histogram with boolean holes only; the thirds ratio exists for A _{zz} alone. Seeded: <code>reco.harmonic_ratio_fit</code> , <code>spin_state_ratio</code> , <code>expected_counts_by_fill</code>	missing → seeded
Full-luminosity pseudo-data	—	<code>sample.phi_histogram_pseudo</code> : exact expected counts per ϕ' bin, Poisson-fluctuated; statistically exact for a binned estimator, blind to reconstruction by construction	exists (statistics only)
Amplitude → Δ	Kinematic conversion with measured F ₂ , R; unfolding	<code>money_delta_extraction.py</code> : model bin-centering K-factor with true kinematics; no migration matrix, model F ₂	partial
Radiative corrections	Unpolarized RC on the yields; tensor RC	None (plans/07 WP4 bound planned)	missing
Coherent: recoil four-vector and t	t from the recoil, x _p from M _X	<code>coherent.recoil_lab</code> (p _T = $\sqrt{ t }$, R, x _L), <code>sample_t</code> ; t never reconstructed; x _p and M _X absent (f _{coh} written against Bjorken x). Seeded: <code>reco.recoil_fourvector</code> (exact light-cone), <code>t_from_fourvectors</code>	partial → seeded
Coherent: Roman-Pot measurement	Transport, divergence smearing, rectangular cutout, angular cut	<code>farforward.route_charged</code> : scalar, ϕ -symmetric p _T cut, no resolution, constant 0.20 GeV at all energies. Seeded: <code>reco.rp_measure</code> , <code>rp_hole_acceptance</code> , <code>tag_pt_cut</code>	partial → seeded
Coherent: exclusivity and vetoes	OMD/ZDC/B0 vetoes, rapidity gap, Z-ID, t fit	<code>coherent.veto_table</code> bookkeeping only; no event-level breakup model (BeAGLE blocked on FLUKA)	missing
Coherent: two-azimuth harmonic analysis	R(α , β) per (x _p , t, Q ²) bin	Single ϕ' (electron) with the anchored a ₂ of the recoil azimuth; no 2-D structure	missing
Full-simulation far-forward	Geant4 acceptance, ElCrecon transport	<code>tools/fullsim</code> gun scans (hit level, RP at 32.5/34.3 m); no ⁶ Li-nucleus gun, no reconstruction	partial

Verdict. The simulation is a validated *truth-level* modulation machine with exact statistics: the physics kernel, the spin-density bookkeeping and the binned estimator are correct and tested, and the money plots are statistically faithful projections. What it did not contain is the measurement layer: no four-vectors, no frames, no kinematic-reconstruction method, no hadronic final state, no reconstructed azimuth, no spin-state-sorted ratio, no Roman-Pot measurement model. Two of the missing pieces matter for the headline numbers rather than for their decoration — the hadronic-method requirement at the $y \approx 0.01$ sweet spots (which changes the binning and the energy strategy, not the reach) and the estimator (which changes the error by 0.67× and removes an uncontrolled systematic) — and one changes the coherent case (the angular cut and the cutout geometry). None of them touches the generator kernel; all of them belong to plans/07 WP3.

5.1 • What a minimal realistic chain looks like in polligen

With the seeded module the reconstructed-level pseudo-experiment of WP3 is a composition, not a new design. For the inclusive channel: (i) sample spin-labelled events from `InclusiveSampler` for the m = ± 1 -rich and m = 0-rich categories of a two-fill run plan (`bookkeeping` already supports it); (ii) build k' with `reco.electron_fourvector` and a uniform ϕ_e , transform to the lab and back, smear with `smear_electron` (calorimeter energy, track angles) and weight with $\epsilon_{\text{elD}}(\eta)$ and a smooth $\epsilon(\phi')$; (iii) reconstruct Q² with `electron_method` and x with `mixed_method` from a parametrized `hadronic_y`; (iv) compute ϕ' with

`azimuth_wrt_lepton_plane` ; (v) histogram per spin state in reconstructed bins and fit with `harmonic_ratio_fit` ; (vi) correct with the migration matrix of the same sample and convert with the model F2; (vii) close against the truth amplitudes of `effective_modulation` — the reconstructed-level analogue of the existing generator-level gate. For the coherent channel: sample t , ϕ_t , x_p ; build P' with `recoil_fourvector` ; emulate the pots with `rp_measure` (rectangular cutout, angular cut at the beam's p_u); compute α and β covariantly; inject a_t , a_e and the unpolarized $u_{1,2}$; fit the 2-D ratio. The hadronic final state itself, the breakup backgrounds and the real acceptance maps remain Phase-2 items and are the honest limit of "simple detector effects".

6 • Recommendations

1. Re-derive money plots 5 and 7 in reconstructed (x , Q^2) bins with the mixed method and a 15–30% hadronic- y band; quote the electron-method-only variant at $y \geq 0.05$ as the fallback, and use the low-energy configuration for the $x \approx 0.1$ bins.
2. Replace the single-fill fit by the spin-state-sorted ratio with the $m = \pm 1$ -rich / $m = 0$ -rich pattern in all projections; update δA to $2\sqrt{(2/N)/(P_+ - P_0)}$ ($0.67\times$ the current values at the same source purity) and drop the "self-normalized within a single fill" wording in favour of "acceptance and relative luminosity cancel in the spin-state ratio".
3. Fit $\cos 2\phi'$ and $\sin 2\phi'$ together; report the $\sin 2\phi'$ coefficient as the spin-axis calibration.
4. In the coherent channel, present the anchored deformation term as a modulation of the recoil azimuth, add the two-azimuth harmonic fit to the analysis description, replace the constant 0.20 GeV cut by the angular cut $10\sigma_\theta \cdot 6p_u$ in `coherent.py` (curves versus σ_θ and p_u , plans/07 WP5), and state the cutout geometry as an explicit assumption to be replaced by the ePIC design.
5. Resolve the a_n normalization convention of the deuteron anchor at the source before the amplitudes are quoted again.
6. Ask the ePIC far-forward and inclusive working groups for two numbers the simulation cannot produce: the Σ -method $\delta y/y$ at $y = 0.01$ – 0.05 for $e +$ light ions, and the Roman-Pot cutout dimensions (β^* , emittance, pot geometry) for the light-ion optics.

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(73 and 149 μrad); the lithium optics are undocumented (plans/04 #11). Hadronic-method resolutions are assumptions (15–30%), not ePIC results. Built by `reports/build_report.py` ; figures from `evgen/scripts/reco_chain_figures.py` ; display mathematics typeset at build time.